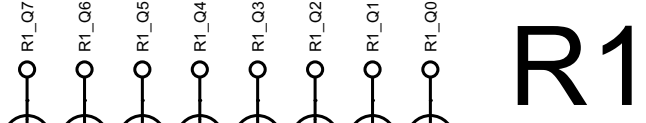
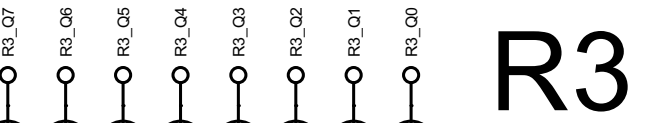
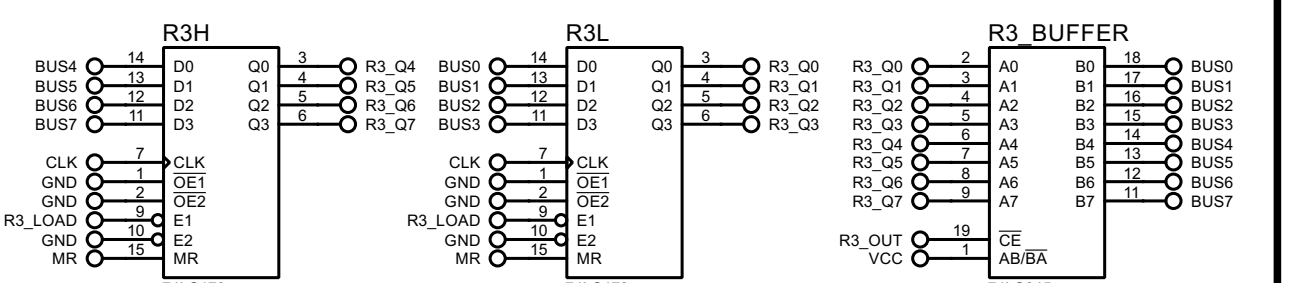
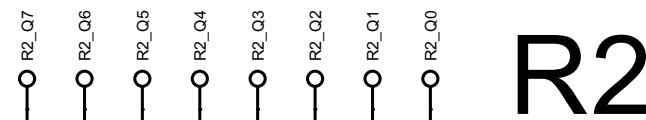
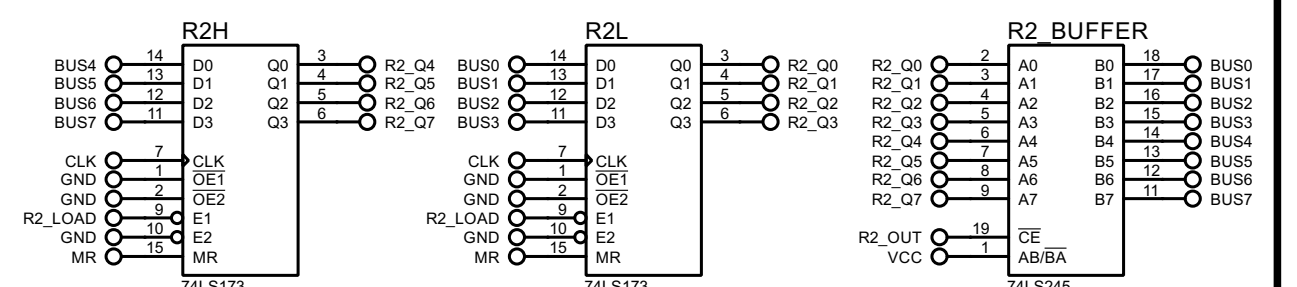


- Add Register B to Register A ($RA = RA + RB$)
- Shift Register A Right by 1 bit ($RA = RA \gg 1$)
- Shift Register A Left by 1 bit ($RA = RA \ll 1$)
- Invert Register A ($RA = \neg RA$)
- AND Register A and B ($RA = RA \wedge RB$)
- OR Register A and B ($RA = RA \vee RB$)
- XOR Register A and B ($RA = RA \wedge RB$)
- Compare RA and RB (Updates flags, RA is unchanged)
- Load data from RAM address into Register A (2 bytes)
- [r] Store Register A data to RAM address (2 bytes)
 - Load immediate 8-bit value into Register A (2 bytes)
 - Jump to address (2 bytes)
 - Jump to address if Carry flag is 1 (2 bytes)
 - Jump to address if Zero/Equal flag is 1 (2 bytes)
 - Clear all ALU flags (Carry and Zero = 0)
- CPU execution (Stops the clock)



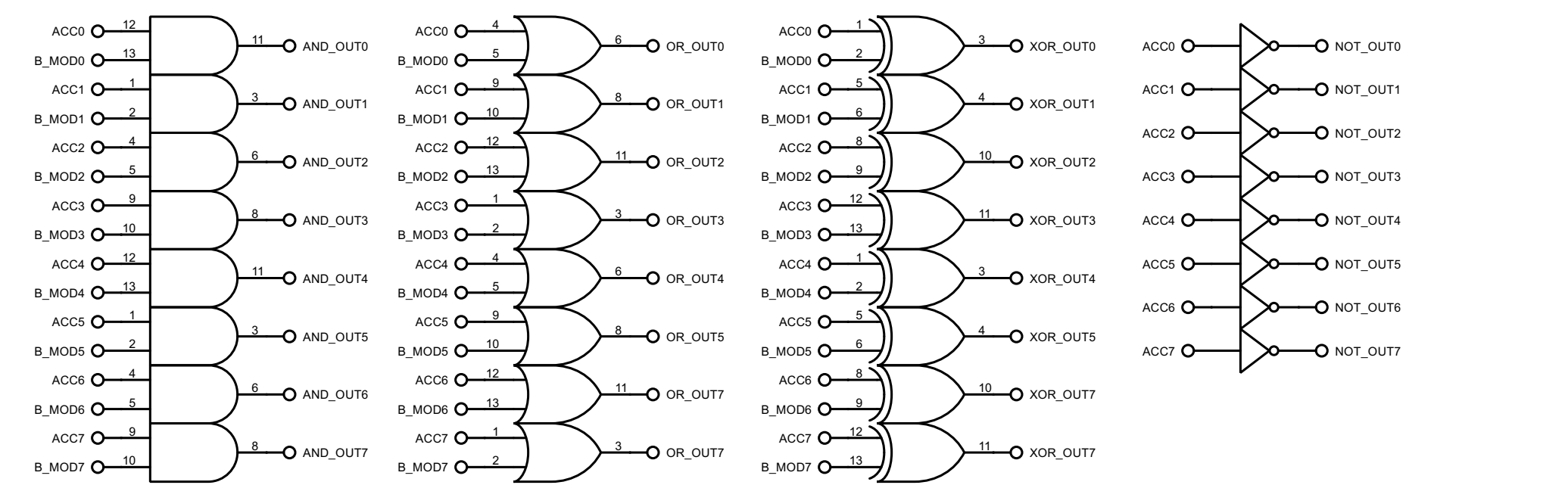
REGISTERS



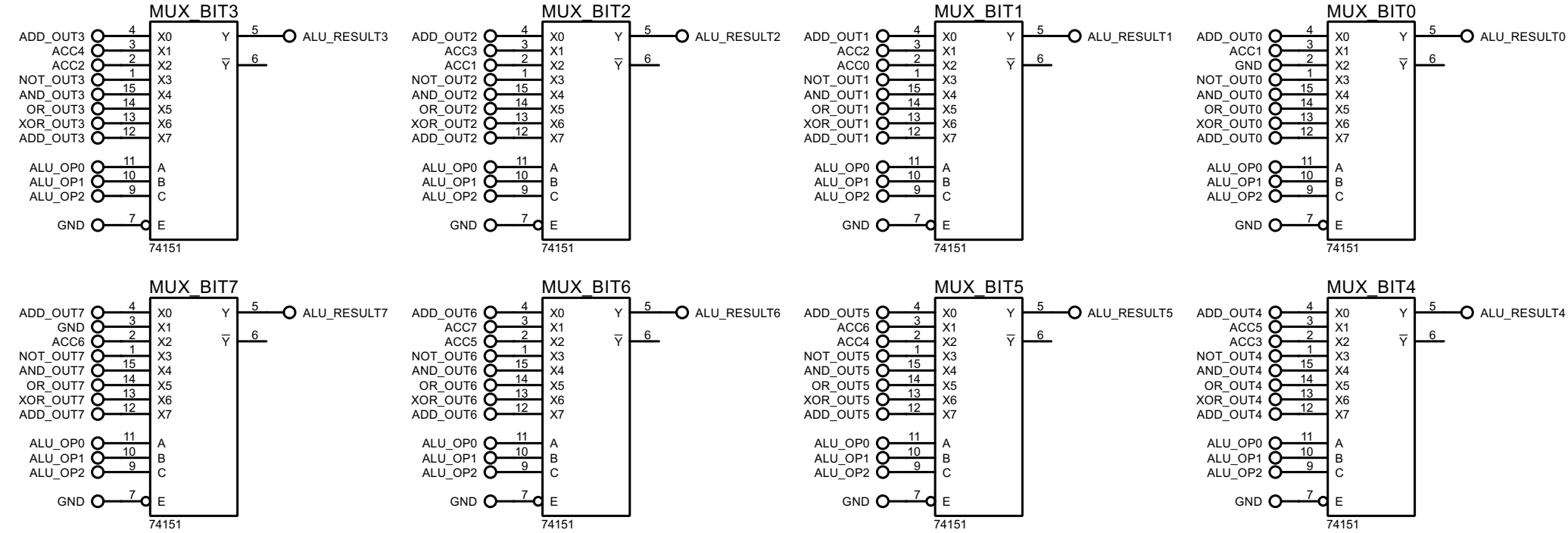
ALU



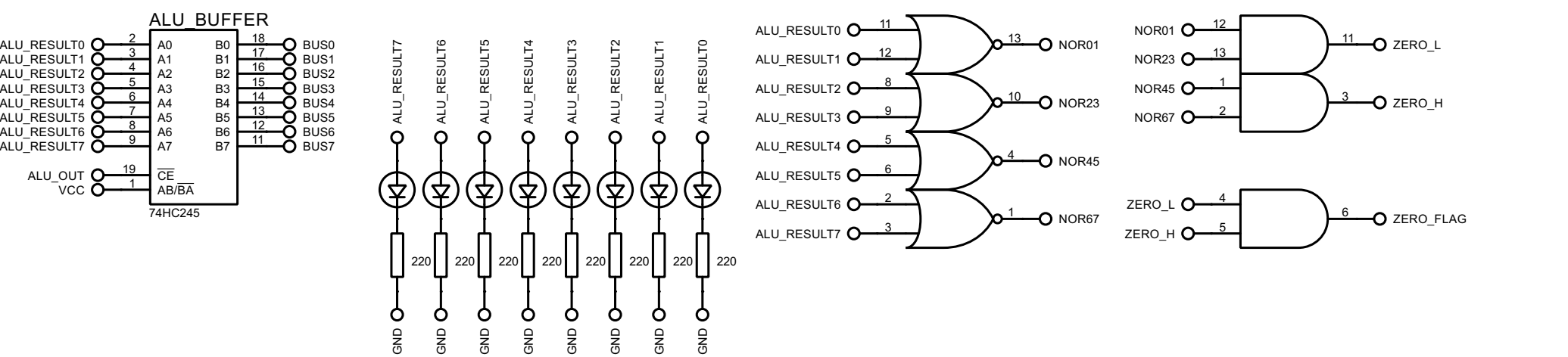
Logical Operation (AND, OR, XOR, NOT)



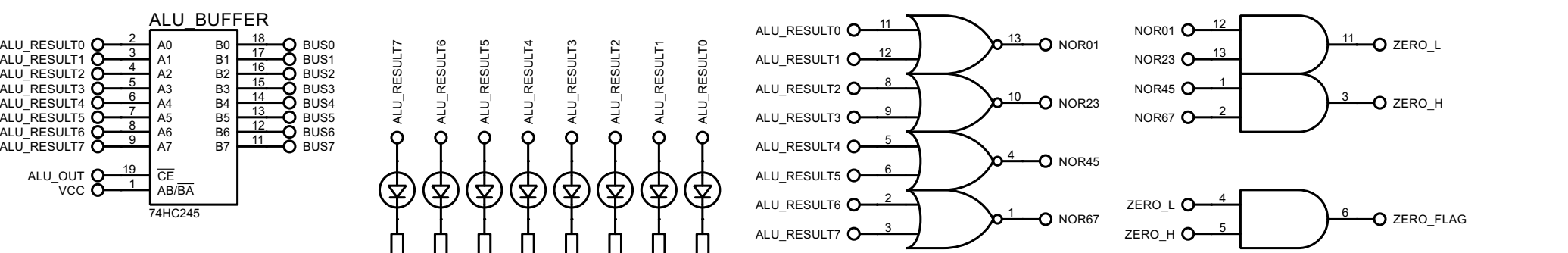
Operating SHR/SHL and multiplex the ALU result



ALU Result



Z and E Flag Logic



Flag Register

